

2) Find limit points, if exist(s), for each of the following:

a) $(4, 3, 2, 1, 4, 3, 2, 1, \dots)$

b) $(3, 2, 2.5, 2.5, 2.333\dots, 2.666\dots, 2.25, 2.75, \dots)$

$= (2+1, 3-1, 2+1/2, 3-1/2, 2+1/3, 3-1/3, 2+1/4, 3-1/4, \dots)$

.....

3) Show that $\lim_{n \rightarrow \infty} n^{(1/n)} = 1$

.....

4) Show that the sequence (n) , for all natural numbers n , does not have a limit

.....

13.4 PROPERTIES OF SETS OF REAL NUMBERS

We have discussed the concepts of open *interval*, e. g. $]2, 4[= \{x: 2 < x < 4\}$; closed *interval* e. g. $[2, 4] = \{x: 2 \leq x \leq 4\}$; semi-open/semi-closed *interval* e. g. $]2, 4] = \{x: 2 < x \leq 4\}$ etc.

Open interval is open in the sense that we cannot tell its boundary elements, e. g. in $]2, 4[$, we do not know the first/least member of $]2, 4[$, because $2 \notin]2, 4[$, and immediately next greater number is not known. A *closed interval* is closed in the sense, we know its boundary points/numbers e. g. 2 and 4 in $[2, 4]$.

The concepts of open and closed *intervals* have been extended to open and closed *sets* as follows.

13.4.1 Definitions: Open Sets, Closed Sets, Bounded Sets, Compact Sets

A) **Open set:** A *set* S of real numbers is said to be **open**, if for each element a of S , there is a number $\varepsilon > 0$ such that the open interval $]a - \varepsilon, a + \varepsilon[\subseteq S$. In other words, every member of S is *well inside* S , and hence, no member is a boundary point.

Example 13.18

i) **Every open interval $A =]a, b[$ is an open set.** Let $x \in A$, then $a < x < b$, and $(x - a) > 0$, $(b - x) > 0$. Let $\varepsilon = \min. ((x - a), (b - x)) > 0$. Then, $]x - \varepsilon, x + \varepsilon[\subseteq]a, b[$.

ii) **Union of open intervals is an open set:**

We consider union of two open intervals. But the argument is valid for any number of open intervals. Let $A =]a_1, b_1[\cup]a_2, b_2[$ be union of two open intervals. We may assume $b_1 < a_2$, otherwise either the two intervals collapse into one interval, or the two intervals may be mutually interchanged. Let $x \in A$, then $x \in]a_1, b_1[$ or $x \in]a_2, b_2[$, say $x \in]a_2, b_2[$ then as above $\exists \varepsilon > 0$. Such that $]x - \varepsilon, x + \varepsilon[\subseteq]a_2, b_2[\subseteq A$. Hence the proof.

B) Closed set: A set $S \subseteq \mathbb{R}$ is said to be a closed if $\mathbb{R} - S = (-\infty, +\infty) - S$ (the set of real numbers not in S) is open. $\mathbb{R}$, the set of real numbers is also denoted as open interval $(-\infty, +\infty)$.

Example 13.19

- i) **Every closed interval $A = [a, b]$ is a closed set.** For a closed interval $[a, b]$, we have $\mathbb{R} - [a, b] = (-\infty, a) \cup (b, +\infty)$, but each of $(-\infty, a)$, and $(b, +\infty)$ is an open interval, hence, their union is an open set.
- ii) Every finite set $\{a_1, a_2, \dots, a_n\}$ is closed. We prove the result for 3-element set $\{a_1, a_2, a_3\}$ which easily may be extended to n -element set. Consider $\mathbb{R} - \{a_1, a_2, a_3\} = (-\infty, a_1) \cup (a_1, a_2) \cup (a_2, +\infty)$. But each of three intervals is open, hence, each is an open set, and union of open set sets is open. Thus, $\mathbb{R} - \{a_1, a_2, a_3\}$ is open, Therefore, $\{a_1, a_2, a_3\}$ is closed.

Remarks: The two sets, and only the following two sets, are both closed and open: (i) $\emptyset$, the empty set, and (ii) $\mathbb{R}$, the set of all real numbers.

C) Bounded set: The concept of *bounded set* generalizes the concept of *bounded sequence*. A set S is said to be **bounded** if it is both (i) bounded above, and (ii) bounded below. (i) A set S is **bounded above**, if there is a real number M , such that $x \leq M$ for all $x \in S$ (ii) A set S is **bounded below**, if there is a real number M' , such that, $M' \leq x$, for all $x \in S$.

Example 13.20

- 1) Each of
 - i) $]3, 10^{10}[$,
 - ii) $[-10^{10}, 1000]$, and
 - iii) $\{1, 2, 10, 10000\} \cup [-10^{10}, 1000]$ is bounded.

Upper and lower Bounds respectively are

- i) 2 and 10^{11} , among other bounds
 - ii) -10^{10} and 1000 among others
 - iii) -10^{10} and 10000 among others
- 2) None of (i) $(-\infty, 10)$, and (ii) $(100, \infty)$ is bounded.
 - i) is not bounded below, but is bounded above with one of upper bounds being 10,

- ii) is not bounded above, but is bounded below, with one of the lower bounds being 100.

D) **Compact set:** A subset of $\mathbb{R}$ is called **compact**, if it is both (i) closed, and (ii) bounded.

Example 13.21

Each of (i) $[3, 10^{10}]$, (ii) $[-10^{10}, 1000]$, and (iii) $\{1, 2, 10, 10000\} \cup [-10^{10}, 1000]$ is bounded and closed. Hence, each is compact.

13.4.2 Bolzano-Weierstrass Theorem and its Economic Applications

Bolzano-Weierstrass Theorem for sets says that every infinite and bounded set has a limit point.

The concept of limit point of a *sequence* is generalized to the concept of limit point of a *set*. A real number p is called a limit point of *set* S of real numbers if for every $\varepsilon > 0$ the open interval $[p - \varepsilon, p + \varepsilon]$ contains a point of S different from p .

Theorem: A point p is a limit point of a set S if and only if for every $\varepsilon > 0$ the open interval $[p - \varepsilon, p + \varepsilon]$ contains infinitely many points of S .

Limit point of a set is also called accumulation/condensation/ cluster point of a set.

Example 13.22

- For an interval $]a, b[$, $[a, b]$, $]a, b]$, $[a, b[$, whether open, closed, semi-open, every point of it is its limit point.
- For an open interval $]a, b[$, or semi-open interval $]a, b]$, or $[a, b[$, both boundary points viz. a and b are also its limit points.

There are different important equilibrium concepts in economics, the proofs of the existence of which often require variations of the Bolzano–Weierstrass theorem. One example is the existence of a Pareto efficient allocation. The Bolzano–Weierstrass theorem allows one to prove that if the set of allocations is compact and non-empty, then the system has a Pareto-efficient allocation. (Wikipedia)

Check Your Progress 4

- Prove each of the following:
 - A closed *interval* is not an open *set*.
 - A semi-closed/semi-open *interval* is not an open *set*.
 - None of (i) $(-\infty, 10)$, and (ii) $(100, \infty)$ is bounded.
 - Every finite set $\{a_1, a_2, \dots, a_n\}$ is compact.

.....

2) Find limit points of the following sets:

a) $\{1, 2, 3, \dots\}$

b) $(3, 5] \cup \{6, 7\} \cup [8, 9]$

.....

13.5 ANALYSIS OF SEVERAL VARIABLES

So far, we have discussed various concepts like sequence, limit, continuity, derivative, integral etc. *in R , the set of real numbers*. In the rest of this unit, we extend/ generalize some of these concepts to R^n *the set of n -tuples of real numbers*. We start with the concept of '*function of several variables*'.

13.5.1 Introduction to Multivariable Functions

We initiate our discussion of multivariable functions with some examples (i) from everyday life of the use of multivariable functions, and then about (ii) how to evaluate such functions for a set of values of the involved variables.

Example 13.23

- i) The area (A) of a rectangle depends on its length (L) and breadth (B), and it is given by the following well-known equation

$$A = L \times B.$$

Thus, the (dependent) variable A depends on two (independent) variables, and hence, A is a function of two variables, viz. L and B .

- ii) The amount of money (M) that becomes due, from a bank, depends on (i) the demand deposit (i.e., money deposited) (DD), (ii) rate of annual interest (r), and (iii) time (t) in years over which the money is deposited. M is given by the following well-known equation.

$$M = DD (1 + r)^t$$

Thus, M depends on three variables, and hence, M is a function of three variables, viz. DD , r and t .

We know from our previous understanding that, for plotting a graph, evaluation of functional value corresponding to independent variable is required. The same is true for plotting graphs of functions of more than one variable. Next, we consider some examples of functional evaluation of more than one variable.

- iii) Find the domain, range and the value of the function given by

$$f(x, y) = 2x^3y + 2y^2 - 3x,$$

at each of the points (i) $x = -2$, and $y = 3$, and (ii) $x = 3$, and $y = -2$.

Solution: As the function is defined for every pair of values of (x, y) , therefore, Domain of f is $R \times R$.

Also, as every real number u can be assumed as a value of f (because, for $x = u/(-3)$ and $y = 0$, $f(x, y) = u$.) Hence, range of f is R

$$(a) \quad f(-2, 3) = 2(-2)^3(3) + 2(3)^2 - 3(-2) = -48 + 18 + 6 = 24$$

$$(b) \quad f(3, -2) = 2(3)^3(-2) + 2(-2)^2 - 3(3) = -108 + 8 - 9 = -109.$$

13.5.2 R^n , the n -dimensional real space & n -dimensional Euclidean Space

In Unit 3, we discussed the concept of cross-product of two sets as follows.

Cross-product of two sets X and Y , for $X \neq \emptyset \neq Y$, denoted by $X \times Y$, is the set of all ordered pairs (x, y) with x belonging to X and y belonging to Y , i. e.

$$X \times Y = \{(x, y): x \in X, y \in Y\}.$$

The concept of cross-product can be generalized to *cross-product of n non-empty sets*, with n being any natural number. We consider the special case—when each of the n involved sets is R , the set of real numbers—as defined below.

$R^n = \{(a_1, a_2, \dots, a_n): a_i \in R\}$. The elements of R^n are called n -tuples

We may introduce *algebraic structures on R^n* , similar to the structures on R , by defining operations like

For $\underline{x} = (x_1, x_2, \dots, x_n)$, and $\underline{y} = (y_1, y_2, \dots, y_n)$ in R^n , and $r \in R$, we define

$$\underline{x} + \underline{y} = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n), \quad \dots (13.3)$$

$$\text{and, } r \underline{x} = (rx_1, rx_2, \dots, rx_n). \quad \dots (13.4)$$

$$\underline{0} = (0, 0, \dots, 0),$$

$$\text{and } -\underline{x} = (-x_1, -x_2, \dots, -x_n).$$

Also, we know from coordinate geometry and 3-dimensional geometry, the metric/topological concepts of (*Euclidean*) *distance* (denoted by d below) between two points, and the *length/norm* of straight-line, or a vector (denoted by $\|\underline{x}\|$, or sometime by $|\underline{x}|$). First, we discuss these concepts only for $\underline{x} = (x_1, x_2) \in R^2$.

$$d(\underline{x}, \underline{y}) = d((x_1, x_2), (y_1, y_2)) = \sqrt{[(x_1 - y_1)^2 + (x_2 - y_2)^2]}, \quad \text{and}$$

$$\|\underline{x}\| = d((x_1, x_2), (0, 0)) = \sqrt{[(x_1)^2 + (x_2)^2]} = |\underline{x}|.$$

These metric/ topological concepts are extended to R^n as follows.

For $\underline{x} = (x_1, x_2, \dots, x_n)$, and $\underline{y} = (y_1, y_2, \dots, y_n)$ in R^n

i) Euclidean Distance

$$d(\underline{x}, \underline{y}) = d((x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n)) = \sqrt{[(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2]} \quad \dots (13.5)$$

ii) Norm

$$\|\underline{x}\| = d((x_1, x_2, \dots, x_n), (0, 0, \dots, 0)) = \sqrt{[(x_1)^2 + (x_2)^2 + \dots + (x_n)^2]} = |\underline{x}| \quad \dots (13.6)$$

$$\text{As } \underline{x} - \underline{y} = (x_1 - y_1, x_2 - y_2, \dots, x_n - y_n); \|\underline{x} - \underline{y}\| = \sqrt{[(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2]}$$

$$\text{Hence, } \|\underline{x} - \underline{y}\| = d(\underline{x}, \underline{y}) \quad \dots (13.7)$$

13.5.3 Functions over Euclidean Spaces

Like functions with domain as $\mathbb{R}$ and/ or codomain as $\mathbb{R}$, we can define functions with domain as $\mathbb{R}^n$ and/or codomain as $\mathbb{R}^n$, for different natural numbers n . For example, (13.3) to (13.6) of 13.5.2 above may be considered as functions respectively as follows:

For $\underline{x} = (x_1, x_2, \dots, x_n)$, and $\underline{y} = (y_1, y_2, \dots, y_n)$ in $\mathbb{R}^n$, and $r \in \mathbb{R}$,

i) $+: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, such that

$$+: ((x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n)) = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n), \quad \dots (13.8)$$

ii) Product by a scalar ‘.’ is defined as

$\mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, such that

$$r\underline{x} = r(x_1, x_2, \dots, x_n) = (rx_1, rx_2, \dots, rx_n), \quad \dots (13.9)$$

where rx_i is the product of two real numbers r and x_i .

Equations (13.5) and (13.6) of 13.5.2 give rise to the following functions

i) $d: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$, such that

$$d(\underline{x}, \underline{y}) = d((x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n)) = \sqrt{[(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2]} \quad \dots (13.10)$$

ii) $\|\cdot\|: \mathbb{R}^n \rightarrow \mathbb{R}$, such that

$$\|\underline{x}\| = d((x_1, x_2, \dots, x_n), (0, 0, \dots, 0)) = \sqrt{[(x_1)^2 + (x_2)^2 + \dots + (x_n)^2]} \quad \dots (13.11)$$

iii) Another well-known function, with domain/ codomain involving $\mathbb{R}^n$ defined as follows, is called *Scalar/dot Product*

Scalar: $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$, such that

$$\text{Scalar}((x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n)) = (x_1, x_2, \dots, x_n) \cdot (y_1, y_2, \dots, y_n) = x_1 y_1 + x_2 y_2 + \dots + x_n y_n,$$

where $(x_i y_i)$ is the product of two real numbers x_i and y_i .

vi) Projection functions from $\mathbb{R}^k$ to $\mathbb{R}^m$, for $m < k$, are explained through the following examples

Let $k = 4$, $m = 3$. Then, we have the following projection functions

$$R_{1,2,3}: \mathbb{R}^4 \rightarrow \mathbb{R}^3 \text{ with}$$

$$R_{1,2,3}((x_1, x_2, x_3, x_4)) \rightarrow (x_1, x_2, x_3);$$

$R_{2,3,4}: R^4 \rightarrow R^3$ with

$R_{2,3,4}((x_1, x_2, x_3, x_4)) \rightarrow (x_2, x_3, x_4)$ etc.

Let $k = 5, m = 2$. Then, we have the following projection functions

$R_{1,2}: R^5 \rightarrow R^2$ with

$R_{1,2}((x_1, x_2, x_3, x_4, x_5)) \rightarrow (x_1, x_2)$;

$R_{3,5}: R^5 \rightarrow R^2$ with

$R_{3,5}((x_1, x_2, x_3, x_4, x_5)) \rightarrow (x_3, x_5)$;

etc.

Example 13.24

For each of the two 5-tuples (i) $\underline{x} = (2, -1, -3, 4, -2)$ and (ii) $\underline{y} = (3, -1, -3, 2, -3)$, find

- the norm $\| \cdot \|$ for each
- Euclidean distance between the two
- scalar product of the two

Solution:

$$\begin{aligned} \text{i) } \|\underline{x}\| &= \|(2, -1, -3, 4, -2)\| = \sqrt{[(2)^2 + (-1)^2 + (-3)^2 + (4)^2 + (-2)^2]} \\ &= \sqrt{[4 + 1 + 9 + 16 + 4]} = \sqrt{34}. \end{aligned}$$

$$\begin{aligned} \|\underline{y}\| &= \|(3, -1, -3, 2, -3)\| = \sqrt{[(3)^2 + (-1)^2 + (-3)^2 + (2)^2 + (-3)^2]} \\ &= \sqrt{[9 + 1 + 9 + 4 + 9]} = \sqrt{32} \end{aligned}$$

- Euclidean distance between

$$\underline{x} = (2, -1, -3, 4, -2), \text{ and } \underline{y} = (3, -1, -3, 2, -3)$$

$$\begin{aligned} \text{is } d(\underline{x}, \underline{y}) &= \sqrt{[(2-3)^2 + (-1-(-1))^2 + (-3-(-3))^2 + (4-2)^2 + (-2-(-3))^2]} \\ &= \sqrt{[1 + 0 + 0 + 4 + 1]} = \sqrt{6}. \end{aligned}$$

- Scalar product of $\underline{x} = (2, -1, -3, 4, -2)$ and $\underline{y} = (3, -1, -3, 2, -3)$ is given by

$$\begin{aligned} \underline{x} \cdot \underline{y} &= (2)(3) + (-1)(-1) + (-3)(-3) + 4(2) + (-2)(-3) = 6 + 1 + 9 + 8 \\ &+ 6 = 30. \end{aligned}$$

Check Your Progress 5

- Find the domain, range and the value of the function

$$f(x, y, z) = 2xy^2z + 2y^3z^2 - 3xz$$

at each of the points

- $x = -1$, and $y = -2$, and $z = 2$

- $x = 1$, and $y = 3$, and $z = -2$

.....

- 2) For each of the two 5-tuples (a) $\underline{x} = (4, 2, -1, -2, -3)$ and (b) $\underline{y} = (-1, 3, -3, -3, 2)$, find
- the norm $\|\cdot\|$ for each
 - Euclidean distance between the two
 - scalar product of the two

.....

13.5.4 Concepts of Analysis of $\mathbb{R}^n$

Many, but not all, of the concepts—already discussed for $\mathbb{R}$, the set of real numbers—like limit, continuity, derivation/derivability, integration/integrability, sequence of real numbers, theorems like Bolzano-Weierstrass for real numbers have been *extended to $\mathbb{R}^n$* , the Euclidean Space of n -dimension. The conceptual extensions, which have been achieved, use some of the following ideas:

- For a sequence $(s_1, s_2, \dots, s_i, \dots, s_n)$ in $\mathbb{R}$, we have $s_i \in \mathbb{R}$, whereas for a sequence $(s_1, s_2, \dots, s_i, \dots, s_n)$ in $\mathbb{R}^n$, we have $s_i \in \mathbb{R}^n$, i. e., $s_i = (s_{i1}, s_{i2}, \dots, s_{in})$, where $s_{ij} \in \mathbb{R}$.
- For discussing the concepts of limit, continuity, derivability etc. in $\mathbb{R}^n$, we make the following modifications:
 - instead of $x, y \in \mathbb{R}$, we take $\underline{x}, \underline{y} \in \mathbb{R}^n$, where $\underline{x} = (x_1, x_2, \dots, x_n)$ and $\underline{y} = (y_1, y_2, \dots, y_n)$, with $x_i, y_i \in \mathbb{R}$.
 - for the concepts of size and distance—instead of absolute value function $|x|$, and distance function $|x-a|$ respectively in $\mathbb{R}$ —we use respectively
 - the norm $\|\underline{x}\| = \sqrt{[(x_1)^2 + (x_2)^2 + \dots + (x_n)^2]}$, and
 - $d: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$, such that

$$d(\underline{x}, \underline{y}) = d(\underline{x}, \underline{y}) = \sqrt{[(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2]},$$

$$\text{where, } \underline{x} = (x_1, x_2, \dots, x_n) \text{ and } \underline{y} = (y_1, y_2, \dots, y_n).$$

In the above, we must note that each of $\underline{x}$ and $\underline{y}$ is just a single point/member of $\mathbb{R}^n$.

Now, we define some of the concepts and state some theorems. First, we Generalization of the concepts of ‘open set’, ‘closed set’ etc. to $\mathbb{R}^n$. Then we discuss the concepts for only *real-valued functions* f , i.e., for $f: \mathbb{R}^n \rightarrow \mathbb{R}$.

Afterwards we generalize these concepts to functions $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, for some natural number m .

and state the relevant theorems only for this type of functions.

A) Generalization of the concepts of 'open set', 'closed set' etc. to $\mathbb{R}^n$

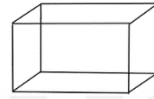
To simplify the discussion, the topic is first discussed for $\mathbb{R}^2$, and then generalized to $\mathbb{R}^n$.

In $\mathbb{R}^2$, the concept of *open interval* is generalized to *open rectangular*, and preferably to *open disc* as follows:

Open Rectangle in $\mathbb{R}^2 = \{(x_1, x_2): a_1 < x_1 < b_1, \text{ and } a_2 < x_2 < b_2 \text{ with } a_1 < b_1, \text{ and } a_2 < b_2\}$,



In general, Open Rectangle in $\mathbb{R}^n = \{\underline{x} = (x_1, x_2, \dots, x_n): \text{for some fixed } \underline{a} = (a_1, a_2, \dots, a_n), \underline{b} = (b_1, b_2, \dots, b_n) \text{ with } a_i < x_i < b_i, \text{ with } i = 1, 2, \dots, n\}$,



a sort of cuboid without boundary point.

But in our discussion, we will use only the concept of open disc, defined as follows.

Open disc in $\mathbb{R}^2 = \{(x_1, x_2): a_i - \varepsilon < x_i < a_i + \varepsilon, \text{ for } \varepsilon > 0, \text{ for some fixed } (a_1, a_2) \in \mathbb{R}^2\}$,

In general, in $\mathbb{R}^n$

Open Disc of radius ε centered at $\underline{a}$

$= \{\underline{x} = (x_1, x_2, \dots, x_n): \text{for some fixed } \underline{a} = (a_1, a_2, \dots, a_n), \|\underline{x} - \underline{a}\| < \varepsilon, \text{ for } \varepsilon > 0\}$,

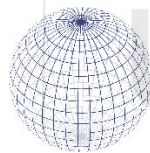


Figure 13.1

a sort of sphere, *without boundary points*, in hyper-space of n-dimension.

Similarly, the concept of *closed interval* is generalized to

closed disc $= \{\underline{x} = (x_1, x_2, \dots, x_n): \text{for some fixed } \underline{a} = (a_1, a_2, \dots, a_n), \|\underline{x} - \underline{a}\| \leq \varepsilon, \text{ for } \varepsilon > 0\}$,

a sort of sphere, *including the boundary points*, in hyper-space of n-dimension.

The concepts of open set, closed set are defined as follows

Open set in $\mathbb{R}^n$: A set S of in $\mathbb{R}^n$ is said to be *open*, if for each element $\underline{a}$ of $S \subseteq \mathbb{R}^n$, there is an *Open Disc* of radius $\varepsilon \subseteq S$. In other words, every member of S is *well inside* S , and hence, no member is a boundary point.

Closed set in $\mathbb{R}^n$: A set $S \subseteq \mathbb{R}^n$ is said to be a closed if $\mathbb{R}^n - S$ (= the set of members of $\mathbb{R}^n$ which are not in S) is open.

Bounded Subset of $\mathbb{R}^n$: A subset S of $\mathbb{R}^n$ is said to be *bounded* if it is both (i) bounded above, and (ii) bounded below. (i) A set S is *bounded above*, if there is a real number M , such that $x_i \leq M$ for all $\underline{x} = (x_1, x_2, \dots, x_n) \in S$ and for $1 \leq i \leq n$. (ii) A set S is *bounded below*, if there is a real number M' , such that $M' \leq x_i$, for all $\underline{x} = (x_1, x_2, \dots, x_n) \in S$ and for $1 \leq i \leq n$.

Compact set: A subset of $\mathbb{R}^n$ is called *compact*, if it is both (i) closed, and (ii) bounded.

Limit point of a subset of $\mathbb{R}^n$: A point $\underline{p} = (p_1, p_2, \dots, p_n)$ of $\mathbb{R}^n$ is called a limit point of subset S of $\mathbb{R}^n$, if every open disc with *radius* $\varepsilon > 0$ contains a point of S different from $\underline{p}$. **The Limit point of a set is also called accumulation/condensation/ cluster point of the set.**

B) Limit and continuity of a real-valued function, i.e., function $f: \mathbb{R}^n \rightarrow \mathbb{R}$

For a real-valued function $f: D \rightarrow \mathbb{R}$, where $D \subseteq \mathbb{R}^n$ and the point $\underline{a} = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$, we say

$$\lim_{\underline{x} \rightarrow \underline{a}} f(\underline{x})$$

$$\underline{x} \rightarrow \underline{a}$$

exists and equals the real number l if and only if for given $\varepsilon > 0$, $\exists \delta > 0$, such that whenever $\|\underline{x} - \underline{a}\| < \delta$, then $|f(\underline{x}) - l| < \varepsilon$

(Note: both $f(\underline{x})$ and l are only real numbers, not members of $\mathbb{R}^n$)

Continuity of a function at a point of $\mathbb{R}^n$

A real-valued function $f: D \rightarrow \mathbb{R}$, where $D \subseteq \mathbb{R}^n$ and the point $\underline{a} = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$, we say f is continuous at $\underline{x} = \underline{a}$ if

$$\text{i) } \lim_{\underline{x} \rightarrow \underline{a}} f(\underline{x}) \text{ exists}$$

$$\underline{x} \rightarrow \underline{a}$$

and

$$\text{ii) } \lim_{\underline{x} \rightarrow \underline{a}} f(\underline{x}) = f(\underline{a}).$$

$$\underline{x} \rightarrow \underline{a}$$

C) Limit and continuity of a function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$

The only difference in the statements of the definitions of the concepts here and those defined in previous subsection, is that the modulus function $|x|$, for x , a real number, is replaced by the norm function $\|\underline{x}\|$ for $\underline{x} \in \mathbb{R}^m$.

For a function $f: D \rightarrow \mathbb{R}^m$, where $D \subseteq \mathbb{R}^n$ and the point $\underline{a} = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$, we say

$$\lim_{\underline{x} \rightarrow \underline{a}} f(\underline{x})$$

$$\underline{x} \rightarrow \underline{a}$$

exists and equals the $\underline{l} \in \mathbb{R}^m$, if and only if for given $\varepsilon > 0$, $\exists \delta > 0$, such that